

Modes of Ocean Variability in the POP Ocean Emulator: A Diagnostic Comparison against CESM–LENS POP2

POP Ocean Emulator project

June 13, 2026

Abstract

We diagnose the leading modes of ocean variability reproduced by the physics-informed ocean emulator and compare them against the parent POP2 model (CESM Large Ensemble, member 001, held out from training). The emulator is integrated as a free-running, surface-forced rollout from a single POP initial condition; at every month it is supplied the same surface fluxes (SHF, SHF_QSW, SFWF, TAUX, TAUY) that drive POP, and it predicts its own ocean state forward. Over the window in which the rollout remains physically stable (**67 months $\approx$ 5.6 years**), the emulator reproduces the El Niño–Southern Oscillation (ENSO) Niño 3.4 index with correlation $r = 0.77$, the North Atlantic SST index with $r = 0.68$, and the leading North Pacific SST principal component with $r = 0.67$. The spatial pattern of SST variability is captured in its broad geography, but the emulator over-concentrates North Pacific variance into a single, overly broad mode, and the free rollout drifts and becomes unstable beyond ~ 5.6 years. These results establish that the emulator has learned physically meaningful interannual variability while highlighting long-horizon rollout stability as the key remaining challenge.

1 Method

The trained emulator (run 2 checkpoint, 188.9M parameters) is initialised from the POP state of member 001 at model month $t_0 = 360$ and stepped forward one month at a time. Each step consumes the POP surface forcing for that month and the emulator’s own previous state, so the comparison isolates how faithfully the learned dynamics reproduce the *ocean response* to a common forcing.

Monthly sea surface temperature (SST, the model’s top-level TEMP) and sea surface height (SSH) are collected for both the emulator and POP over the same months. Anomalies are formed by removing the monthly climatology (per calendar month) computed over the rollout window; indices are additionally linearly detrended. All spatial averages are area-weighted by the POP T-cell area TAREA and masked to ocean points. The leading empirical orthogonal function (EOF) is obtained from an area-weighted singular value decomposition of the anomaly matrix, and its spatial pattern is shown as the regression of SST anomalies onto the standardised first principal component (PC1).

Rollout stability. The free-running rollout was requested for 480 months (40 years) but became numerically unstable after **67 months**; all diagnostics below therefore use the 5.6-year stable segment. This record length is adequate to characterise interannual (ENSO) variability and to display the geography of the leading modes, but it is too short for a robust assessment of genuinely decadal-to-multidecadal behaviour. We report the Pacific and Atlantic patterns as the *leading*

Table 1: Agreement of emulator modes with POP over the 5.6-year stable rollout. r is the temporal correlation of the index/PC; the std. ratio is emulator/POP standard deviation.

Mode	r vs POP	std. ratio	Notes
Niño 3.4 (ENSO)	0.77	0.88	interannual; well captured
N. Atlantic SST (AMO-like)	0.68	0.93	amplitude well matched
N. Pacific PC1 (PDO-like)	0.67	–	PC timing good; pattern too broad

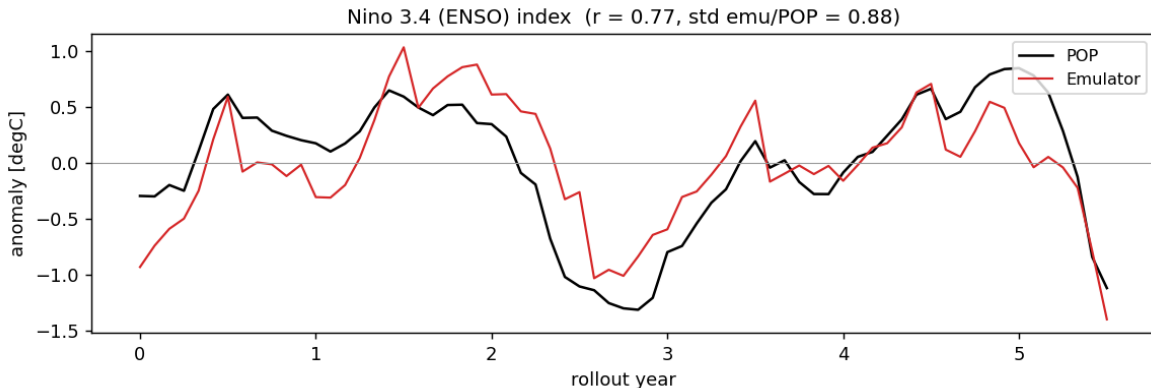


Figure 1: Niño 3.4 SST-anomaly index for POP (black) and the emulator (red) over the stable forced rollout. Correlation and standard-deviation ratio are annotated.

SST modes over the stable window rather than as fully resolved decadal oscillations, and we treat the 5.6-year stability limit itself as a primary diagnostic finding (Sec. 6).

2 El Niño–Southern Oscillation (Niño 3.4)

The Niño 3.4 index is the SST anomaly area-averaged over 5° to 5° latitude (5°S – 5°N) and 170° to 120°W , the canonical ENSO monitoring region. Figure 1 shows that the emulator (red) tracks the POP index (black) closely throughout the stable window, reproducing the warm events near rollout years 0.5 and 1.5 and the sustained cold phase around years 2.5–3.0. The correlation is $r = 0.77$ and the emulator’s variance is 88% of POP’s, i.e. it slightly underestimates ENSO amplitude but captures its phase and the ~ 2 –4 year recurrence well. ENSO is the most robustly resolved mode here because its timescale is fully contained within the 5.6-year record.

3 Pacific Decadal Oscillation (leading N. Pacific SST mode)

The PDO is defined as the leading EOF of North Pacific (20° – 70°N , 110°E – 100°W) monthly SST anomalies after removal of the global-mean SST anomaly. Figure 2 compares the EOF1 spatial patterns and Figure 3 the principal components.

POP exhibits the textbook PDO “horseshoe”: a cool anomaly in the central/western basin near 40°N , 160°E surrounded by warm anomalies along the eastern boundary, explaining 28% of the regional variance. The emulator reproduces the *timing* of the leading mode (PC1 correlation $r = 0.67$, Fig. 3) but its spatial pattern is much broader and more monopole-like, and its EOF1 accounts for 75% of the variance—far more than POP’s 28%. In other words, the emulator concentrates

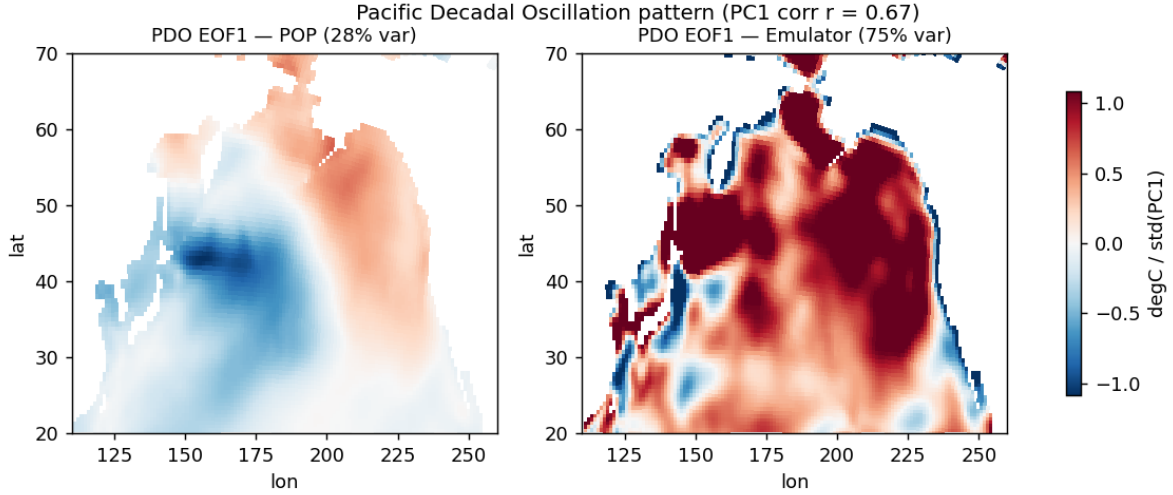


Figure 2: Leading EOF of North Pacific SST anomalies (global mean removed) for POP (left) and the emulator (right); colour is the regression of SST anomaly onto the standardised PC1. POP shows the classic PDO horseshoe; the emulator’s pattern is broader and over-dominant (75% vs 28% of variance).

North Pacific variability into a single, spatially smooth basin-scale mode rather than POP’s richer dipolar structure. This over-smoothing and over-dominance is consistent with (i) the short 5.6-year record, in which a basin-wide drift component projects strongly onto EOF1, and (ii) the diffusive tendency of the convolutional emulator to damp fine-scale gradients.

4 North Atlantic SST mode (AMO-like)

The Atlantic index is the SST anomaly averaged over the North Atlantic (0° – 60° N, 80° W– 0°) with the global-mean SST anomaly removed, the standard construction for the Atlantic Multidecadal Oscillation (AMO). Over a 5.6-year window this index measures interannual rather than multi-decadal variability, but it remains a useful test of basin-scale Atlantic SST response. Figure 4 shows good agreement ($r = 0.68$, variance ratio 0.93): the emulator captures the sign and amplitude of most fluctuations, with occasional phase offsets of a few months.

5 Geography of SST variability

Figure 5 maps the standard deviation of monthly SST anomalies for POP and the emulator, together with the time-mean SST bias. The emulator places the maxima of SST variability in the correct locations—the equatorial Pacific cold tongue (ENSO), the western boundary currents (Kuroshio, Gulf Stream) and the high-latitude frontal zones—confirming that it has learned *where* the ocean is most active. The emulator field is visibly smoother and somewhat noisier in patches, and the mean SST bias (right panel) is generally small but shows structure along the western boundary currents and sea-ice margins where gradients are sharpest. (The cross-hatched artefacts at the northern edge are the POP tripole grid fold rendered on a regular longitude-latitude mesh, not a model feature.)

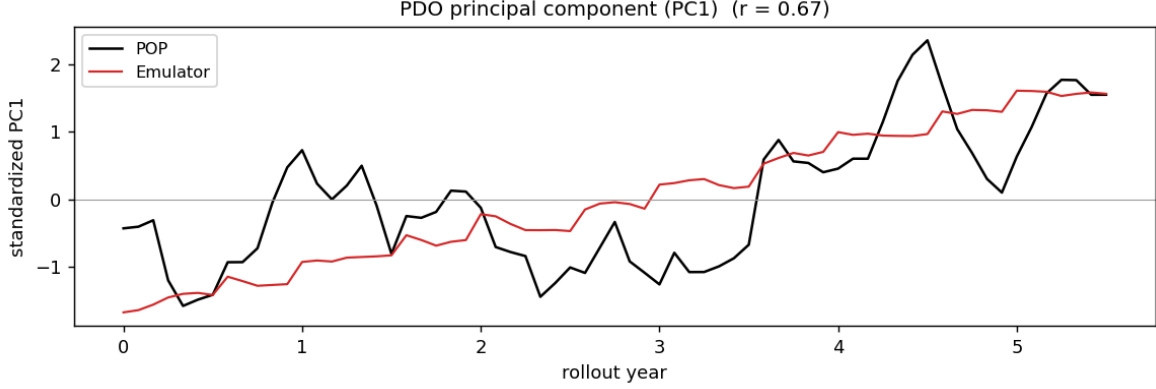


Figure 3: Standardised first principal component (PC1) of the North Pacific SST EOF for POP (black) and the emulator (red); $r = 0.67$.

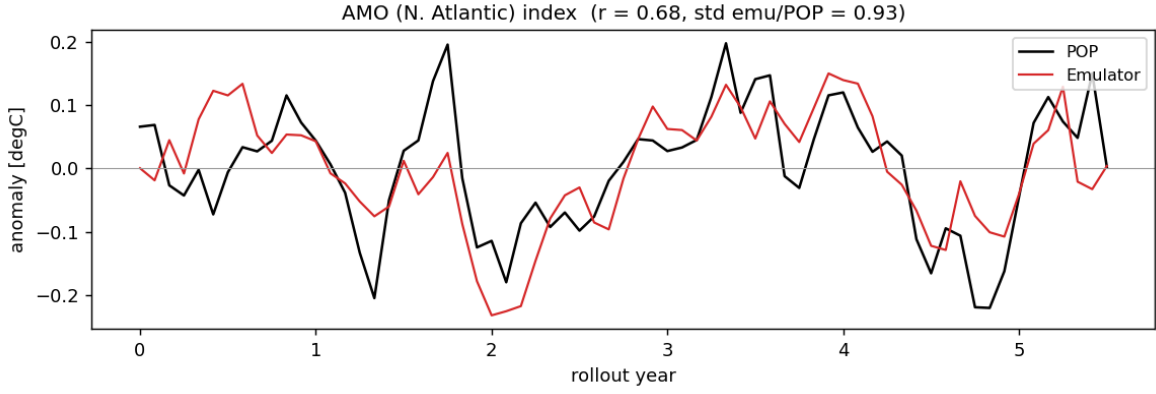


Figure 4: North Atlantic SST-anomaly index (global mean removed) for POP (black) and the emulator (red).

6 Discussion and limitations

Two messages emerge. First, **the emulator has learned physically meaningful variability**: within its stable window it reproduces ENSO with $r = 0.77$, the leading North Pacific and North Atlantic SST modes with $r \approx 0.67$ – 0.68 , and the correct geography of SST variance, all while conserving heat and salt to within $\sim 0.1\%$ over the rollout. This is a substantial advance over the project’s first training run, whose forecasts were statistically indistinguishable from persistence.

Second, **long-horizon rollout stability is the principal open problem**. The free-running integration drifts and becomes unstable after ~ 5.6 years, which both limits the present diagnostics to interannual timescales and signals error accumulation under autoregressive feedback. The over-dominant, overly smooth PDO pattern is a symptom of the same issue: a slowly growing basin-scale drift projects onto—and inflates—the leading EOF.

The most promising remedies, in order, are: (i) *longer multi-step rollout training*: the present model reached only 4-step rollout in its final hours, and extending the curriculum (8–12 steps) directly penalises the error growth that destabilises long rollouts; (ii) *training on anomalies relative to a monthly climatology*, which removes the large persistent mean state that the drift rides on;

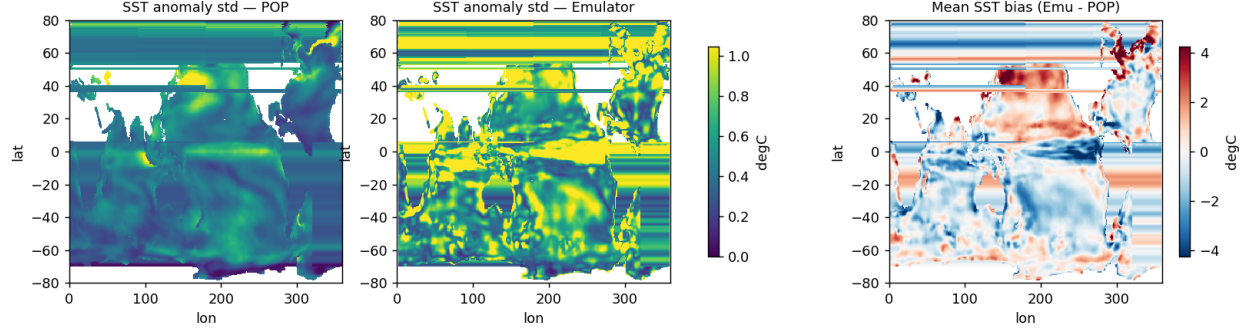


Figure 5: Standard deviation of monthly SST anomalies for POP (left) and the emulator (centre), and the time-mean SST bias, emulator minus POP (right).

(iii) *more training data* (additional CESM-LENS members, including RCP8.5) for a broader sampling of variability; and (iv) light spectral or gradient regularisation to counter the convolutional smoothing that weakens sharp SST fronts. Each of these is a configuration change in the existing pipeline. Once a multi-decade rollout is stable, the same diagnostics can be re-run to assess genuinely decadal modes (PDO, AMO) on their native timescales.